

LLM-powered Gaussian Splatting in VR interactions

Haotian Mao*

Shanghai Jiao Tong University

Yule Quan§

Shanghai Jiao Tong University

Zhuoxiong Xu†

Shanghai Jiao Tong University

Nianchen Deng¶

Shanghai AI Lab

Siyue Wei‡

Shanghai Jiao Tong University

Xubo Yang||

Shanghai Jiao Tong University

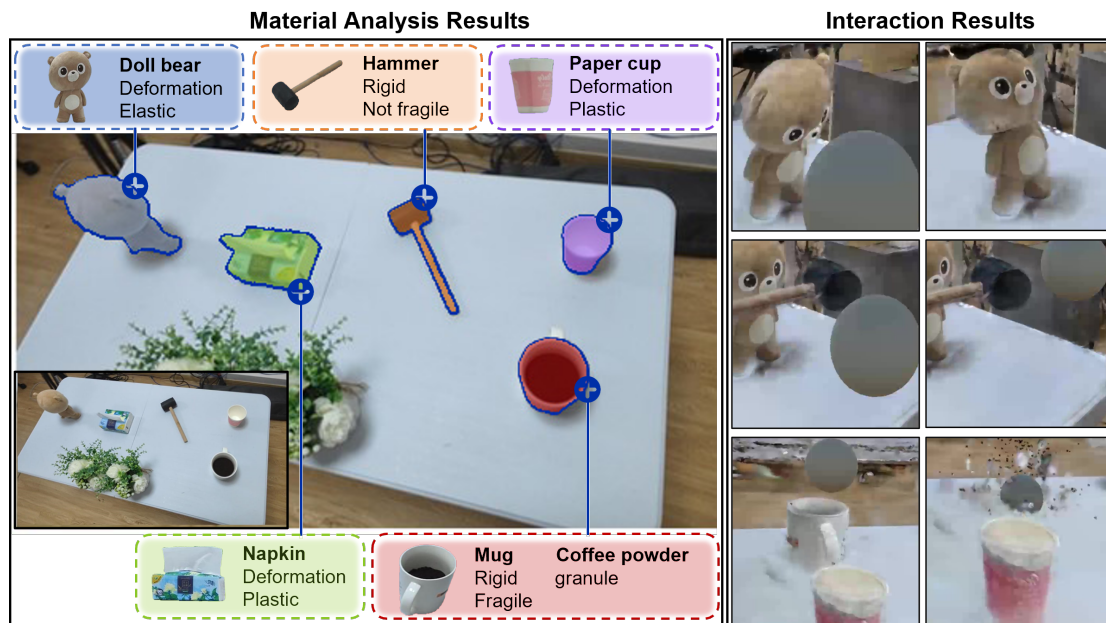


Figure 1: Our system reconstructs the scene with extra features and segments target objects. Crucially, our GPT-powered system can analyze the physical properties of objects, thus enabling simulations that align with real-world phenomena. For instance, when balls are thrown at the doll, it swings; when thrown at the mug, it breaks; and when hitting the hammer, the ball bounces.

ABSTRACT

Recent advances in radiance field rendering, particularly 3D Gaussian Splatting (3DGS), have demonstrated significant potential for VR content creation, offering both high-quality rendering and an efficient production pipeline. However, current physics-based interaction systems for 3DGS are limited to either simplistic, unrealistic simulations or require substantial user input for complex scenes, largely due to the lack of scene comprehension. In this demonstration, we present a highly realistic interactive VR system powered by large language models (LLMs). After object-aware GS reconstruction, we prompt GPT-4o to analyze the physical properties of objects in the scene, which then guide physical simulations that adhere to real-world phenomena. Additionally, We design a GPT-assisted GS inpainting module to complete the areas occluded by manipulated objects. To facilitate rich interaction, we introduce a computationally efficient physical simulation framework through a

PBD-based unified interpolation method, which supports various forms of physical interactions. In our research demonstrations, we reconstruct varieties of scenes enhanced by LLM’s understanding, showcasing how our VR system can support complex, realistic interactions without additional manual design or annotation.

Index Terms: 3D Gaussian Splatting, virtual reality, material analysis

1 INTRODUCTION

With the swift advancement of virtual and augmented reality technologies, increasingly realistic and reliable immersion has emerged, leading to numerous applications in education, healthcare, and assembly training. In addition to traditional modeling techniques, radiance field rendering has demonstrated significant potential in VR asset creation due to its remarkable novel synthesis capabilities. With rapid evolution, 3D Gaussian Splatting (3DGS) [4] has quickly taken precedence for its real-time efficiency and seamless integration with the modern rasterization pipeline. Although existing methods such as VR-GS [2] successfully implement interactive physics-based VR systems based on 3DGS, their method limits interactions to elastic deformation and requires professional knowledge. Other approaches like Physdreamer [10] attempt to predict physical properties, but they are constrained by preconceived assumptions and struggle with complex real-world scenes. Currently, there is no VR system capable of analyzing physical properties and enabling real-time interaction across diverse materials, especially

*e-mail: maohaotian@sjtu.edu.cn

†e-mail: vector-02@sjtu.edu.cn

‡e-mail: weisiyue071024@sjtu.edu.cn

§e-mail: qyl728@sjtu.edu.cn

¶e-mail: dengnianchen@pjlab.org.cn

||e-mail: yangxubo@sjtu.edu.cn

in real-world environments.

Therefore, we introduce an interactive VR system based on 3DGS, which aims to reconstruct complex real-world scenes and facilitate real-time interactions that align with actual phenomena. Leveraging GPT models, our system exhibits strong scene understanding capabilities and performs physical property analysis tasks, enabling it to determine and analyze the physical properties of objects from minimal input images. During the reconstruction procedure, we import extra features as semantic information, favoring subsequent editing and physical simulations. Inspired by skinning algorithms [3, 5], we implement a unified physical simulation method based on PBD [6] and interpolate to acquire Gaussian kernels for rendering while maintaining high efficiency. In our interaction demos, we implement several different interaction patterns such as grabbing, dragging, and shooting. We evaluate our systems in various datasets containing different materials, revealing the effectiveness of both high-quality environment reconstructions and immersive interactions.

2 SYSTEM DESIGN

Our pipeline consists of a Gaussian training component, a LLM-powered analysis component, and an interaction component. During the reconstruction stage, our system reconstructs Gaussian kernels from multi-view images and analyzes the physical properties of each interactive target. These assets are imported into VR settings and enable real-time interactions.

Gaussian training 3DGS [4] represents a scene using a collection of colored 3D Gaussians and renders them into camera views through splatting-based rasterization. To extract each interactive object, we introduce additional identity features to each kernel and jointly train them with consistent masks generated by DEVA [1]. To eliminate artifacts which are unseen regions and caused by movements of objects, we use 2D inpainting method LaMa [7] in each view and train for other iterations to complete the scenes. Our code is built on Gaussian Grouping [9], where a grouping loss that includes 2D identity loss and 3D regularization loss is introduced to enhance robustness and accuracy.

physical analysis We use GPT-4o for material analysis due to its strong visual perception capabilities. Considering that the environment is important for analysis, we input two images: the first one is the original image and the other is the mask marking the target for analysis. It is first demanded to distinguish physical material types such as deformations or rigid, and then is asked to predict concrete values for related attributes. The model is demanded to output formatted replies such as *json* file, which is strongly connected with our underlying physical engine. We also implement a chat dialogue feature so that users can explicitly specify material types.

physically-based interactions For physical simulation and interaction in VR, we build our underlying physical system based on Obi solver [8] in Unity 2022.3.30. We generate physical particles from object surfaces, which participate in physical calculation but keep invisible during rendering. For each Gaussian kernel, we find the closest m particles and calculate their weights at frame 0, where m is determined by a specific phenomenon. Their attributes can be interpolated from related particles. Obi requires mesh for particle generation. So we use Open3D to construct mesh, which directs the particle generation for each segmented object. We only allow particles adsorbed on the surface for rigid, while voxel generation is turned on to fill the deformation and granular material. When generating particles, we choose a fixed density (particle numbers in each voxel or surface) for all simulations. Significantly, we raise the granular particle density to the upper limit to achieve better simulation results. Besides, the mesh enables collision detection in our system. To better fit the scene, we adopt plane fitting at the supporting plane. We also implement the distance field to achieve efficient detection.

3 INTERACTIVE DEMONSTRATION

Our system is tested on an Intel(R) Core(TM) i9-14900K CPU with 64GB of RAM and an NVIDIA GeForce RTX 4080 SUPER GPU with 16GB of memory. We conduct our VR experiments using a Meta Quest 3 VR headset with controllers. We choose publicly available datasets such as *garden*, *wolf*, and *sofa*. We also collect our own datasets *labdesk*, using a mobile phone. These datasets include various physical properties, which are suitable to test the material analysis ability and interaction in our system. We only demonstrate the results in *labdesk* here while the others can be found in our video.

As shown in Figure 1, this dataset consists of an empty paper cup, a mug with coffee powder, a plastic hammer, a doll bear, and a box of tissues. In this scene, users can engage in various interesting interactions. When they try to drag the toy and the tissue, the toy always recovers, while the tissue may collapse due to their different levels of plasticity. They can also hold the hammer and break the mug, causing the coffee powder to spill out.

We find that GPT-4o can achieve high accuracy in category recognition. Given one or two views with precise masks to indicate the target, it can handle complex situations such as distinguishing between the container and its content. Besides, it can refer to the analysis history in the same scene, which helps maintain relative accuracy. However, predicting absolute values without reference is challenging, which may cause inaccurate analysis, particularly across different scenes.

REFERENCES

- [1] H. K. Cheng, S. W. Oh, B. Price, A. Schwing, and J.-Y. Lee. Tracking anything with decoupled video segmentation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 1316–1326, 2023. 2
- [2] Y. Jiang, C. Yu, T. Xie, X. Li, Y. Feng, H. Wang, M. Li, H. Lau, F. Gao, Y. Yang, et al. Vr-gs: A physical dynamics-aware interactive gaussian splatting system in virtual reality. *arXiv preprint arXiv:2401.16663*, 2024. 1
- [3] L. Kavan, S. Collins, J. Žára, and C. O’Sullivan. Skinning with dual quaternions. In *Proceedings of the 2007 symposium on Interactive 3D graphics and games*, pp. 39–46, 2007. 2
- [4] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4):1–14, 2023. 1, 2
- [5] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. J. Black. Smpl: A skinned multi-person linear model. In *Seminal Graphics Papers: Pushing the Boundaries, Volume 2*, pp. 851–866, 2023. 2
- [6] M. Müller, B. Heidelberger, M. Hennix, and J. Ratcliff. Position based dynamics. *Journal of Visual Communication and Image Representation*, 18(2):109–118, 2007. 2
- [7] R. Suvorov, E. Logacheva, A. Mashikhin, A. Remizova, A. Ashukha, A. Silvestrov, N. Kong, H. Goka, K. Park, and V. Lempitsky. Resolution-robust large mask inpainting with fourier convolutions. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pp. 2149–2159, 2022. 2
- [8] O. S. D. Team. Obi solver. <https://obi.virtualmethodstudio.com/>, 2024. 2
- [9] M. Ye, M. Danelljan, F. Yu, and L. Ke. Gaussian grouping: Segment and edit anything in 3d scenes. *arXiv preprint arXiv:2312.00732*, 2023. 2
- [10] T. Zhang, H.-X. Yu, R. Wu, B. Y. Feng, C. Zheng, N. Snaveley, J. Wu, and W. T. Freeman. Physdreamer: Physics-based interaction with 3d objects via video generation. *arXiv preprint arXiv:2404.13026*, 2024. 1